CloudFront + S3 + Route 53 — PowerShell Setup

Step-by-step PowerShell commands to serve a static frontend with a custom domain over HTTPS

Use backtick (`) for line continuation. JSON payloads are assigned to variables using here-strings (@' ... '@) to avoid quoting issues.

Step 1 — Create the S3 Bucket

```
aws s3api create-bucket `
  --bucket my-app-frontend `
  --region us-east-1
```

Step 2 — Block Public Access

CloudFront will handle access — keep the bucket private.

```
aws s3api put-public-access-block `
  --bucket my-app-frontend `
  --public-access-block-configuration `
    "BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=false,RestrictPublicBucket
s=false"
```

Step 3 — Request an ACM Certificate

Must be in us-east-1 for CloudFront. Note the CertificateArn in the output.

```
aws acm request-certificate `
  --domain-name app.example.com `
  --validation-method DNS `
  --region us-east-1
```

Get the DNS validation record:

```
aws acm describe-certificate `
  --certificate-arn arn:aws:acm:us-east-1:ACCOUNT_ID:certificate/CERT_ID `
  --region us-east-1 `
  --query "Certificate.DomainValidationOptions"
```

Step 4 — Add ACM Validation Record to Route 53

Use the Name and Value from the previous output:

```
$validationRecord = @'
{
  "Changes": [{
    "Action": "CREATE",
```

```
    "ResourceRecordSet": {
      "Name": "_abc123.app.example.com.",
      "Type": "CNAME",
      "TTL": 300,
      "ResourceRecords": [{ "Value": "_xyz456.acm-validations.aws." }]
    }
  }]
}
'@

aws route53 change-resource-record-sets `
  --hosted-zone-id YOUR_ZONE_ID `
  --change-batch $validationRecord
```

Wait for the certificate to be issued:

```
aws acm wait certificate-validated `
  --certificate-arn arn:aws:acm:us-east-1:ACCOUNT_ID:certificate/CERT_ID `
  --region us-east-1
```

Step 5 — Create a CloudFront Origin Access Control

Note the Id from the output — you'll need it in Step 6.

```
$oacConfig = @'
{
  "Name": "my-app-oac",
  "OriginAccessControlOriginType": "s3",
  "SigningBehavior": "always",
  "SigningProtocol": "sigv4",
  "Description": ""
}
'@

aws cloudfront create-origin-access-control `
  --origin-access-control-config $oacConfig
```

Step 6 — Create the CloudFront Distribution

Replace YOUR_OAC_ID, ACCOUNT_ID, and CERT_ID with your values.

```
$distConfig = @'
{
  "CallerReference": "my-app-ref-1",
  "Origins": {
    "Quantity": 1,
    "Items": [{
      "Id": "s3-my-app-frontend",
      "DomainName": "my-app-frontend.s3.amazonaws.com",
      "OriginAccessControlId": "YOUR_OAC_ID",
      "S3OriginConfig": { "OriginAccessIdentity": "" }
    }]
```

```
  },
  "DefaultCacheBehavior": {
    "TargetOriginId": "s3-my-app-frontend",
    "ViewerProtocolPolicy": "redirect-to-https",
    "CachePolicyId": "658327ea-f89d-4fab-a63d-7e88639e58f6",
    "AllowedMethods": { "Quantity": 2, "Items": ["GET", "HEAD"] }
  },
  "CustomErrorResponses": {
    "Quantity": 2,
    "Items": [
      { "ErrorCode": 403, "ResponsePagePath": "/index.html", "ResponseCode": "200", "ErrorCa
chingMinTTL": 0 },
      { "ErrorCode": 404, "ResponsePagePath": "/index.html", "ResponseCode": "200", "ErrorCa
chingMinTTL": 0 }
    ]
  },
  "Aliases": { "Quantity": 1, "Items": ["app.example.com"] },
  "ViewerCertificate": {
    "ACMCertificateArn": "arn:aws:acm:us-east-1:ACCOUNT_ID:certificate/CERT_ID",
    "SSLSupportMethod": "sni-only",
    "MinimumProtocolVersion": "TLSv1.2_2021"
  },
  "DefaultRootObject": "index.html",
  "Enabled": true,
  "Comment": ""
}
'@

aws cloudfront create-distribution `
  --distribution-config $distConfig
```

Note the Id (distribution ID) and DomainName (e.g. d1abc.cloudfront.net) from the output.

Step 7 — Update the S3 Bucket Policy

```
$bucketPolicy = @'
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "Service": "cloudfront.amazonaws.com" },
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::my-app-frontend/*",
    "Condition": {
      "StringEquals": {
        "AWS:SourceArn": "arn:aws:cloudfront::ACCOUNT_ID:distribution/DIST_ID"
      }
    }
  }]
}
'@
```

```
aws s3api put-bucket-policy `
  --bucket my-app-frontend `
  --policy $bucketPolicy
```

Step 8 — Point Route 53 at CloudFront

Z2FDTNDATAQYW2 is CloudFront's fixed hosted zone ID — same for everyone.

```
$dnsRecord = @'
{
  "Changes": [{
    "Action": "CREATE",
    "ResourceRecordSet": {
      "Name": "app.example.com",
      "Type": "A",
      "AliasTarget": {
        "HostedZoneId": "Z2FDTNDATAQYW2",
        "DNSName": "YOUR_CLOUDFRONT_DOMAIN.cloudfront.net",
        "EvaluateTargetHealth": false
      }
    }
  }]
}
'@

aws route53 change-resource-record-sets `
  --hosted-zone-id YOUR_ZONE_ID `
  --change-batch $dnsRecord
```

Step 9 — Deploy Your Files

```
aws s3 sync ./dist s3://my-app-frontend --delete

aws cloudfront create-invalidation `
  --distribution-id YOUR_DIST_ID `
  --paths "/*"
```

Values to replace throughout: YOUR_ACCOUNT_ID, YOUR_CERT_ID, YOUR_OAC_ID, YOUR_DIST_ID, YOUR_ZONE_ID, YOUR_CLOUDFRONT_DOMAIN, your domain name, and your bucket name.